

Forecasting load: validation, naives, horizons

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Where we are

- **UE ISD-1020** “Machine Learning / Deep Learning”
- Monday **28 September 2026** · PUIO **B210** · 09:00–12:30
- Session and material: **Stéphane Rivaud**
- On 21 September: machine failures, leakages, boosting. This morning: **series**

In your jobs

Same question, four terrains of this cohort:

- **Power grid** — tomorrow 18:00, how many MW? (RTE, this lab)
- **Transport** — ridership, validations, trains at 24 h
- **Telecom** — data / voice traffic at peak hour
- **Plant** — cadence, load of a line, not only a failure

Decision, then metric, then model — the protocol does not change.

MAE and MAPE, not accuracy

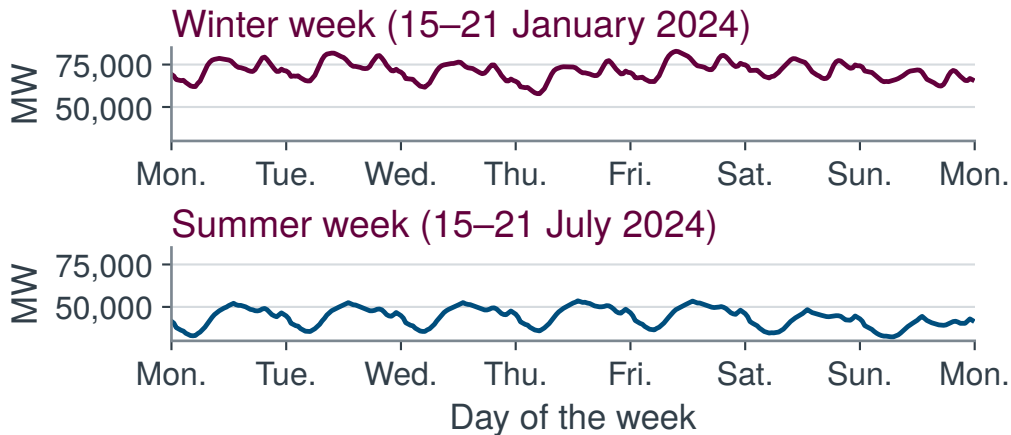
- Target = MW, a **number**, not a class
- **MAE**: mean gap in megawatts
- **MAPE**: the same gap as a percentage of the load
- **21 September** scored **accuracy** on a class. Here: **MAE / MAPE** on MW

What this morning covers

- **Validation**: why a random split **lies** on a series
- **Naive baselines** as the judge ($t - 1$ day, $t - 1$ week)
- **Legal** features at origin t , then a **temporal** cut
- Boosting by **horizon** (1 h, 24 h), then a window $\rightarrow$ MLP
- GRU on the fast track; holiday / heat-wave errors (a **proxy**)

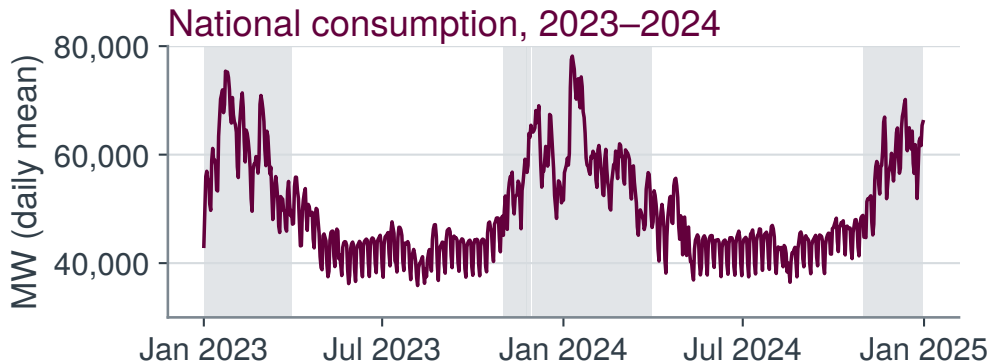
A week of load

éCO2mix, national France, 30 min steps. Same scale, winter vs summer.



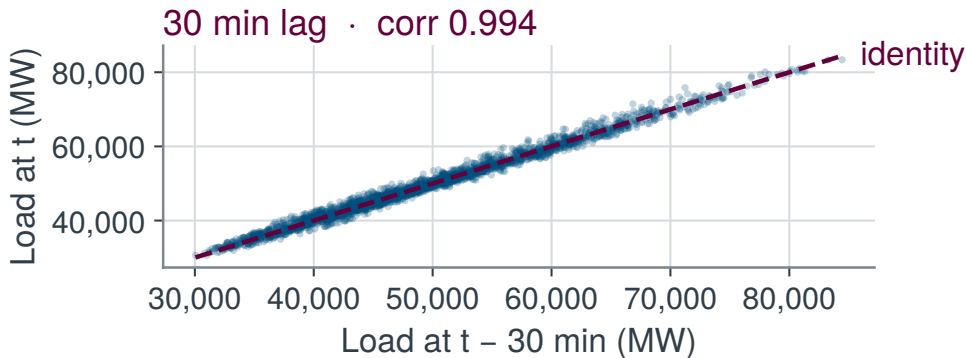
A year of load

Winter high (heating), summer low. Mean **49,912 MW** · min 29,798 · max 85,370. 35,088 points, 731 days.



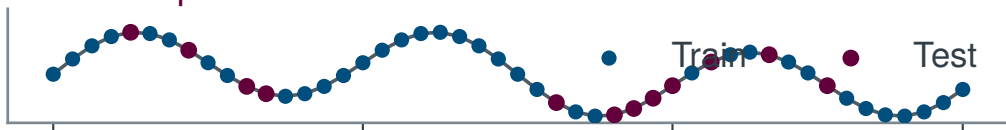
Two neighbouring steps

Short delay: correlation near **one**; day and week copies are far looser. Two adjacent 30 min steps are **almost the same value**.

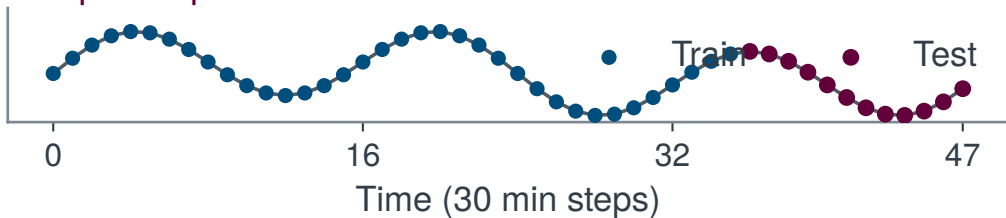


Two ways to cut

Random split: the test is woven into the train



Temporal split: the test is the future



The lie, in MW

Same boosting family, same honest features, **24 h** horizon.

Split	What you see
80/20 shuffled	Looks clearly better
Temporal (past train / future test)	The score you cite

The shuffled split is **roughly a fifth too optimistic**. 2024 looks like 2023: the gap does not explode. It is **real**. We cite the temporal.

Why random CV leaks

- **Most** shuffled test points have their 30 min neighbour **in the train**
- That neighbour is correlated near **one**: the model interpolates, it does not forecast
- The test “sees” **future** and **past** weeks
- Remedy: train origins whose **target** is still in the past; the test is the future

D1 (21 Sep): post-failure diagnostics in $X \rightarrow$ PR-AUC $\sim$ **0.50** $\rightarrow$ **0.945**.

Here the optimism is milder — still **do not publish** shuffled CV.

The naive, the judge

Before boosting, **with no learning**:

- **Persistence:** $y(t + h) \leftarrow y(t)$ (last point seen)
- **Day-naive:** $y(t + h) \leftarrow y(t + h - 1 \text{ day})$ — copying yesterday
- **Week-naive:** $y(t + h) \leftarrow y(t + h - 7 \text{ days})$

At 24 h, persistence and day-naive **coincide**. A model that loses against the **best** of the three has lost — even if it is “deep”.

t-1 day and t-1 week

Same temporal test origins as the model.

Copy	Pattern on this series
Persistence	Strong at 1 h ; coincides with day-naive at 24 h
Day-naive	Hardest judge at 24 h (copying yesterday)
Week-naive	Weaker than D-1 here

Here **D-1 beats D-7**: weather sticks day to day. Beat copying yesterday at 24 h.

Lag features

At origin t , we are only allowed $y(s)$ for $s \leq t$.

- Recent past, yesterday, last week — delays you can defend
- No **centred** mean, no $y(t + 1)$
- A lag is not leakage: it is the past **already observed**

This morning's boosting: lags already observed, plus a **target** calendar you justify.

The calendar is not leakage

Knowing that *tomorrow* is Christmas does not use tomorrow's load.

- Hour, weekday, month, weekend, **target public holiday** — if you include them
- Metropolitan list 2023–2024, written in the notebook (not a package)

Without a holiday bit, holiday errors **blow up** and can lose to copying yesterday.

With it, holidays stay harder than ordinary days, but the flag **saves** Christmas.

Boosting by horizon

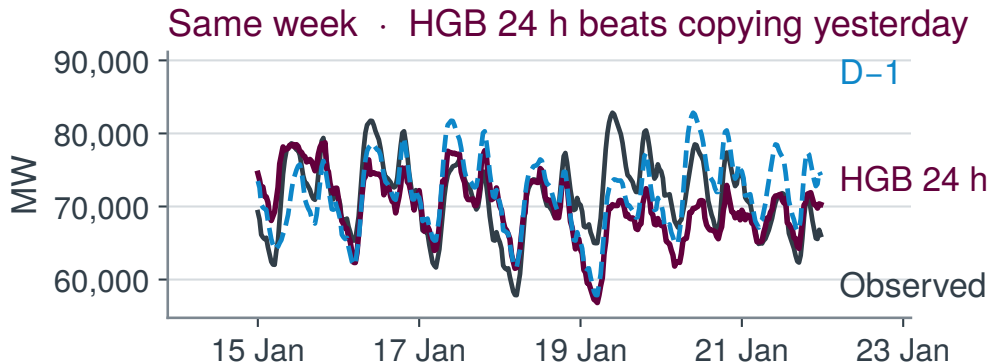
Two models: one for the **1 h** horizon, another for the **24 h** horizon. Not a single model for both.

- Histogram gradient boosting (scikit-learn), modest iteration budget, fixed seed
- Train: targets still in the **past** · test: **future** origins
- Same feature family you chose: target calendar + legal lags

This is the morning **tabular baseline**, to be beaten by a network if it can.

HGB against the naives

Boosting **beats** persistence at 1 h and **beats** copying yesterday at 24 h — measured on the temporal hold-out.



Window → MLP

48 steps = **24 h** of load, **flattened**. No calendar.

$$\hat{y}(t + 1 \text{ h}) = f_{\text{MLP}}(y(t - 47), \dots, y(t))$$

- Small feed-forward net, two hidden ReLU layers, linear head
- MSE, Adam, a dozen epochs, CPU · window and target standardised on the train
- The network must **read the profile** in the window, without the holiday flag

GRU intuition

The MLP ignores **order** once the window is flattened.

- A GRU reads the window **in time**
- A hidden state: what to keep, what to drop (forget / update gates)
- Last state $\rightarrow$ a linear head $\rightarrow \hat{y}$

Fast track: same data as the MLP, fewer epochs. This is not an RNN course.

MLP, GRU, HGB

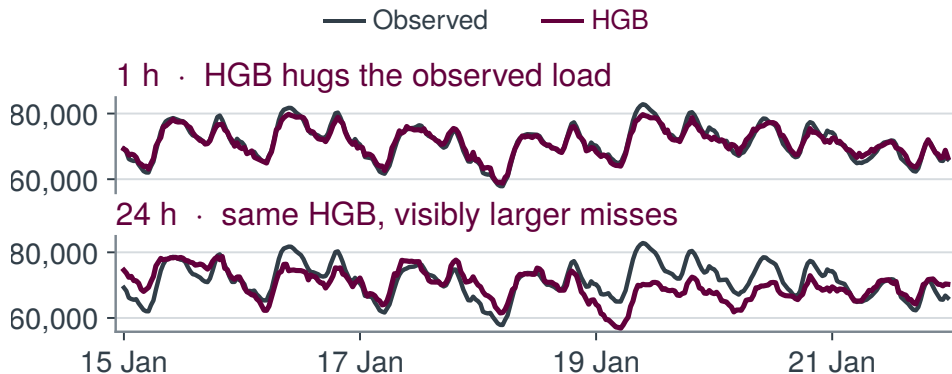
1 h horizon, temporal split, CPU.

Model	vs persistence	vs boosting
Persistence	—	floor
GRU (short CPU run)	usually wins	loses
MLP	wins	loses
HGB	wins	—

The networks **beat** the naive and **lose** against boosting. On *this* series, calendar + lags suffice. Deep is not due.

Evaluate by horizon

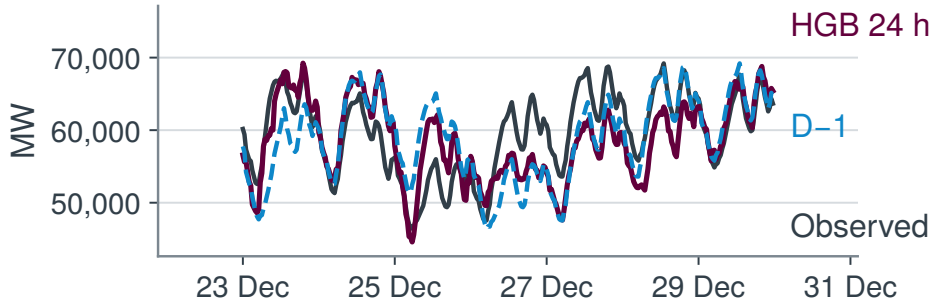
Same model, same week: close at **1 h**, larger misses at **24 h**. A 1 h score does not stand in for a day-ahead brief.



Public holidays

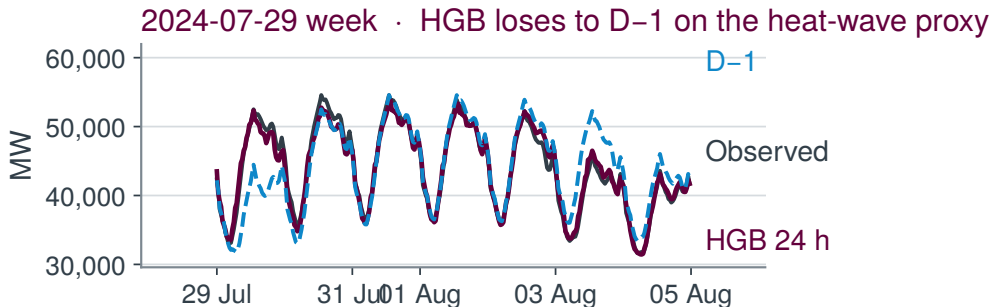
With the flag, boosting **beats** copying yesterday on holiday targets; without it, holidays go bad. Ordinary days stay easier either way.

23–29 December 2024 · Christmas is a public holiday



Heat waves: a proxy

No temperature. JJA high daily means (upper tail, ~19 days in the reference run). Boosting **loses** to copying yesterday on that proxy. Not a Météo-France label.



Toward a recommendation

- **Deploy** an HGB 1 h / 24 h with a temporal cut and a defended target calendar: it beats the naives
- Do **not** publish a shuffled CV (roughly a fifth too optimistic here)
- **Keep** copying yesterday as judge, especially in summer; encode holidays
- Collect a **temperature** before talking about heat waves; the proxy is not enough
- A session MLP/GRU does not replace boosting on this series

This morning's lab

1. Shared course folder · **copy in Drive**
2. Default **autonomous** (guided if the room needs hints)
3. Thread: seasonality → **legal features** → **temporal cut** → the split's lie
→ naives → HGB → MLP
4. Six **protocol** cells at the bottom (oral exam of 5 October)

Fast track: holiday / heat segments, then GRU on the same 48-step window.

Attributions and licences

Session and material: Stéphane Rivaud.

- [éCO2mix](#) · RTE / ODRE · [Licence Ouverte v2.0](#)
- [scikit-learn MOOC \(Inria\)](#) · modules 2 and 7 · [CC BY 4.0](#)
- [PyTorch tutorials](#) · [BSD-3-Clause](#)
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